

Cryptographic Compliance Requirements

REQ-2026-001

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1 Purpose

This document defines the cryptographic requirements for SendCUIEmail to ensure compliance with federal standards for protecting Controlled Unclassified Information (CUI). These requirements establish the minimum security controls that must be implemented and verified.

2 Scope

These requirements apply to:

- File encryption and decryption operations
- Password/key derivation functions
- Random number generation for cryptographic material
- All platforms where SendCUIEmail operates (Windows, macOS, Linux)

3 Applicable Standards

Standard	Description
32 CFR Part 2002	Controlled Unclassified Information Program
FIPS 197	Advanced Encryption Standard (AES)
FIPS 140 2	Security Requirements for Cryptographic Modules
NIST SP 800-132	Recommendation for Password-Based Key Derivation
NIST SP 800-171	Protecting CUI in Nonfederal Systems (reference)
NIST SP 800-53	Security Controls for Federal Systems (reference)
NIST SP 800-38A	Recommendation for Block Cipher Modes of Operation
NIST SP 800-90A	Recommendation for Random Number Generation
NIST SP 800-63B	Digital Identity Guidelines - Authentication

4 Cryptographic Requirements

Requirements are categorized as:

- **[M]** – Mandatory: Must be implemented
- **[R]** – Recommended: Should be implemented where feasible

4.1 Encryption Algorithm Requirements

ID	Requirement
REQ-1.1	[M] The system SHALL use AES (Advanced Encryption Standard) as defined in FIPS 197 for symmetric encryption.
REQ-1.2	[M] The system SHALL use a key length of 256 bits (AES-256) for all encryption operations.
REQ-1.3	[M] The system SHALL use Cipher Block Chaining (CBC) mode or an authenticated encryption mode (GCM, CCM).
REQ-1.4	[M] The system SHALL generate a unique, random Initialization Vector (IV) for each encryption operation.
REQ-1.5	[M] The IV SHALL be at least 128 bits (16 bytes) for AES-CBC mode.
REQ-1.6	[M] The IV SHALL be stored with or transmitted alongside the ciphertext.

4.2 Key Derivation Requirements

ID	Requirement
REQ-2.1	[M] The system SHALL use PBKDF2 (Password-Based Key Derivation Function 2) as defined in NIST SP 800-132 for deriving encryption keys from passwords.
REQ-2.2	[M] The system SHALL use HMAC-SHA256 as the pseudorandom function (PRF) for PBKDF2.
REQ-2.3	[M] The system SHALL use a minimum of 100,000 iterations for PBKDF2 per NIST SP 800-132 recommendations.
REQ-2.4	[M] The system SHALL generate a unique, random salt for each encryption operation.
REQ-2.5	[M] The salt SHALL be at least 128 bits (16 bytes) as recommended by NIST SP 800-132.
REQ-2.6	[M] The salt SHALL be stored with or transmitted alongside the ciphertext.
REQ-2.7	[R] The iteration count SHOULD be increased over time as computing power increases.

4.3 Random Number Generation Requirements

ID	Requirement
REQ-3.1	[M] The system SHALL use a cryptographically secure random number generator (CSPRNG) for generating salts, IVs, and other cryptographic material.
REQ-3.2	[M] On Windows, the system SHALL use the system RNG provided by .NET (RNGCryptoServiceProvider or RandomNumberGenerator).
REQ-3.3	[M] The CSPRNG SHALL meet the requirements of NIST SP 800-90A or equivalent.

4.4 Password Requirements

ID	Requirement
REQ-4.1	[M] The system SHALL enforce a minimum password length of 8 characters.
REQ-4.2	[M] The system SHALL require password confirmation during encryption to prevent typos.
REQ-4.3	[R] The system SHOULD offer automatic passphrase generation using cryptographic RNG.
REQ-4.4	[M] Passwords SHALL NOT be stored or logged by the system.

ID	Requirement
REQ-4.5	[M] Password input SHALL be masked (not displayed on screen) during entry.

4.5 File Format Requirements

ID	Requirement
REQ-5.1	[M] Encrypted files SHALL contain: salt, IV, and ciphertext in a defined format.
REQ-5.2	[M] The file format SHALL be: [16-byte salt][16-byte IV][ciphertext].
REQ-5.3	[M] Encrypted files SHALL use the .Locked extension to indicate encrypted status.
REQ-5.4	[M] The original file extension SHALL be preserved (e.g., document.pdf becomes document.pdf.Locked).

4.6 Platform Requirements

ID	Requirement
REQ-6.1	[M] The system SHALL use platform-provided cryptographic libraries (no custom implementations).
REQ-6.2	[M] On Windows, the system SHALL use .NET Framework cryptographic classes (System.Security.Cryptography).
REQ-6.3	[R] The system SHOULD detect and warn if FIPS mode is not enabled on Windows.
REQ-6.4	[M] The system SHALL NOT require installation of third-party software on recipient systems.

5 Verification

Each requirement SHALL be verified through one or more of the following methods:

Method	Description
Inspection	Code review to verify implementation matches requirement
Test	Automated or manual testing to verify correct behavior
Analysis	Review of design documentation or cryptographic parameters

Verification results are documented in the corresponding VER document.

6 Requirements Traceability

Requirement	Standard	Section
REQ-1.1	FIPS 197	§5 Algorithm Specification
REQ-1.2	FIPS 197	§6.1 Key Length Requirements
REQ-1.3	NIST SP 800-38A	§6.2 The Cipher Block Chaining Mode
REQ-1.4	NIST SP 800-38A	Appendix C Generation of IVs
REQ-1.5	NIST SP 800-38A	§5.3 Initialization Vector
REQ-1.6	NIST SP 800-38A	§5.3 Initialization Vector
REQ-2.1	NIST SP 800-132	§5.3 PBKDF Specification
REQ-2.2	NIST SP 800-132	§5.3 (PRF selection)
REQ-2.3	NIST SP 800-132	§5.1.1.2 Iteration Count
REQ-2.4–2.6	NIST SP 800-132	§5.1 Salt
REQ-3.1–3.3	NIST SP 800-90A	§8.6.7 Get_entropy_input
REQ-4.1–4.2	NIST SP 800-63B	§5.1.1.1 Memorized Secret Authenticators
REQ-4.3	NIST SP 800-63B	§5.1.1.2 Memorized Secret Verifiers
REQ-4.4–4.5	NIST SP 800-63B	§5.1.1.2 (storage and display)
REQ-5.1–5.4	32 CFR 2002	§2002.20 Marking
REQ-6.1–6.2	FIPS 140-2	§4.1 Cryptographic Module Specification
REQ-6.3	FIPS 140-2	§4.9.1 FIPS Mode
REQ-6.4	32 CFR 2002	§2002.16 Accessing and disseminating

7 Revision History

Version	Date	Description
1.0	January 22, 2026	Initial release